CryptIQ

FOUNDER'S CODE 2026
1-Page Proposal

See how hackers would break your customers' data – then lock it for good, before DPDP and quantum arrive.

1. Topic Title: CryptIQ – Crypto-Risk Scanner, Attack Lab & Auto-Fixer

2. Domain: Cybersecurity & Digital Safety

3. Problem Statement

By **13 May 2027**, every Indian business handling personal data must apply **reasonable security safeguards such as encryption and masking**, with penalties of **up to ₹250 crore** for failures (DPDP Rules 2025). Readiness is low: an EY survey found **~81% of organisations** had not updated their frameworks, and MSMEs lag most (Inc42, Jul 2026). Small businesses face two risks they cannot see: **(1) weak protection today** – plaintext phone numbers, MD5 passwords and hard-coded keys that leak instantly in a breach; **(2) "harvest now, decrypt later"** – data protected by RSA/ECC can be stolen now and decrypted once large quantum computers exist; NIST published post-quantum standards (FIPS 203, ML-KEM) in 2024. Existing DPDP tools cover consent and notices; **none shows an owner, in plain language, whether their own code protects data – or lets them watch it being attacked.**

4. Proposed Solution

CryptIQ scans a business's own code, database schema and server settings **offline** – code never leaves their machine – and turns hidden crypto risks into something anyone can see:

1. **Scan** – builds a crypto inventory: where personal data lives, how it is hashed or encrypted, which keys, libraries and TLS versions are used; 0–100 readiness score.
2. **Attack Lab** – shows the leaked customer table a hacker would see. **Crack passwords** runs a real dictionary attack on weak hashes and reveals them. **Quantum attack (Shor's algorithm, simulated)** factors the RSA key protecting stored data – using a deliberately small key we can break live – and decrypts customers' emails and phone numbers.
3. **Fix** – one click generates patches as a pull request the developer reviews: MD5 → Argon2, plaintext → field-level encryption, RSA → hybrid ML-KEM, hard-coded keys → secrets vault.
4. **Prove** – the score only rises after four live checks pass: **re-scan** (critical issues 3 → 0), **re-attack** ("0 of 3,000 cracked", "Shor not applicable – lattice-based key"), **app still works** (automated login and checkout test) and **new data protected** (a new sign-up is stored encrypted). Owner gets an evidence report, a verifiable trust badge and a "verify it yourself" checklist.

5. Technical Approach

Technology Stack: Python scanner (AST + regex rule packs, ~15 rules) · crypto inventory in CycloneDX CBOM format · Attack Lab: dictionary-attack engine, Pollard's-rho factoring of a toy RSA key to simulate Shor, open-source ML-KEM library · FastAPI · PostgreSQL (SQLite for prototype) · React + Tailwind dashboard · offline CLI · GitHub integration · PDF reports.

Key Features: transparent score (100 − 15×critical − 7×high − 3×medium) · data map · owner / developer views · Attack Lab · auto-fix pull requests · 4-check verification · CI guard that blocks new weak crypto · partner dashboard · public badge page.

Prototype / MVP Feasibility: built from scratch during the event in a new repository (no code reused from our earlier crypto research). **Live demo:** "Chennai Organics", an online shop with 3,000 fake customers – a judge signs up with a test password → it is cracked live → **Fix all** → 4 checks pass → score **38 → 91** → the judge signs up again and the same attacks fail. The quantum attack is clearly labelled as a simulation: no quantum computer can break RSA-2048 today.

6. Business Approach

Target Customers / Users: startups and MSMEs holding customer data (e-commerce, clinics, ed-tech, fintech partners), their developers, and the CA and IT-service firms that advise them.

Revenue Model (estimates, to be validated with pilot customers):

Tier	Price	What the customer gets
Free	₹0	Readiness score + top 3 risks + one Attack Lab run – drives adoption
Starter	₹999 / month	One small website or app: all findings, fixes, re-scans, report
Business	₹2,999 / month	Multiple projects, CI guard, evidence report, trust badge
Partner (CA / IT / audit firms)	₹500 / client / month	Multi-client dashboard, white-label reports; partner resells at a margin (e.g. 25 clients ≈ ₹1.5 lakh / year)

Benchmarks: a manual data audit costs ₹1–10 lakh and a fractional DPO ₹2–8 lakh a year (Inc42) – a Business plan is ~₹36,000 a year.

Go-to-Market: free scans through startup incubators and college E-cells → partner with CA and IT-service firms who already serve MSMEs → "Are you DPDP-ready?" campaigns before May 2027 → industry associations.

7. Expected Impact

Small businesses find and fix real data-protection gaps before the DPDP deadline · fewer readable leaks of customers' personal data · compliance evidence at a fraction of consultant cost · an early, affordable path to post-quantum encryption. **Measured by:** score improvement per scan, critical risks fixed, and cost per report vs manual audit.

8. Challenges & Risks

Risk	Mitigation
Attack tools being misused	Attack Lab runs only on the user's own scanned assets or built-in demo data; no external targets
Over-claiming quantum risk	Shor attack labelled as simulation on a small key; RSA-2048 is not breakable today
"Compliance" is legal, not only technical	Positioned as security-safeguard evidence; legal sign-off via partner CA/legal firms
False positives / wrong fixes	Fixes are reviewed pull requests; marked done only after re-scan, re-attack and app tests pass
MSMEs won't pay	Free tier + partner channel; priced far below manual audits

Sources: DPDP Rules 2025 (MeitY, notified 13 Nov 2025) via EY · Inc42 (17 Jul 2026), citing EY India survey · NIST FIPS 203 (ML-KEM), Aug 2024.